ZYNVEX SOLUTIONS

BATCH 5 VIRTUAL INTERNSHIP — OFFICIAL PROJECT PROPOSAL FORM

⚠ **MANDATORY SUBMISSION DEADLINE: SUNDAY AT 4:00 PM**

All Batch 5 virtual interns must complete and submit this formal project proposal document before Sunday, 4:00 PM. Fill out your details, project overview, and divide your project execution into 4 weekly modules starting Monday.

1. INTERN & PROJECT DETAILS

Intern Full Name	**Adeel Abbasi**
Internship ID	**ZYNVEX-CERT-2068**
Program / Track	Frontend Development (Track 01 - UI/UX Focus)
Selected Project Title	**BeatGrid — Interactive Web Audio Sampler & Looper (FED-02)**

2. PROJECT DESCRIPTION (4 LINES)

Problem: Music producers and beginners want to quickly test drum sounds without opening heavy software like FL Studio.
Target Users: Student beatmakers, bedroom producers, and anyone who wants to play simple beats in their web browser.
Tech Stack: HTML5, CSS3, JavaScript, and the browser Web Audio API to make sure sounds play with zero delay.
Deliverables: A responsive 16-pad drum machine with keyboard shortcuts, volume slider, preset kits, and mobile support.

3. INTERN 4-MODULE EXECUTION PLAN (SELF-DIVIDED)

Module & Dates	Intern Work Breakdown & Planned Deliverables
Module 1 Oct 26 – Nov 01	**Project Setup & Pad Layout:** Set up the project with Vite, build the 16 drum pads with clean CSS styling, collect the drum sound files (kicks, snares, hats), and write the basic code to load audio into the browser.
Module 2 Nov 02 – Nov 08	**Sound Playback & Keyboard Controls:** Wire up the pads so sounds play instantly on click, add keyboard event listeners (keys 1-4, Q-R, A-F, Z-V) to trigger beats from PC, and add glowing visual feedback on button press.
Module 3 Nov 09 – Nov 15	**Sound Kits & Volume Controls:** Add a kit selector to switch between different sound styles (808 kit, Lo-Fi kit), add a master volume slider, and save the user's selected kit in local storage.
Module 4 Nov 16 – Nov 22	**Mobile Testing, Polish & Deployment:** Test touch response on mobile screens, check sound playback across Chrome and Safari, write a clear project README on GitHub, and deploy the website live on Vercel.